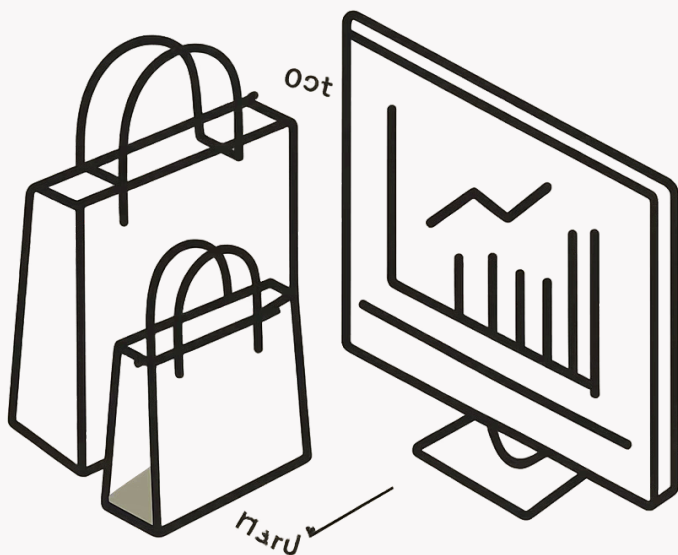




Customer Shopping Behavior Analysis

Transactional data analysis across **3,900 purchases** — uncovering spending patterns, segments, and product preferences to guide strategic decisions.

Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

Demographics

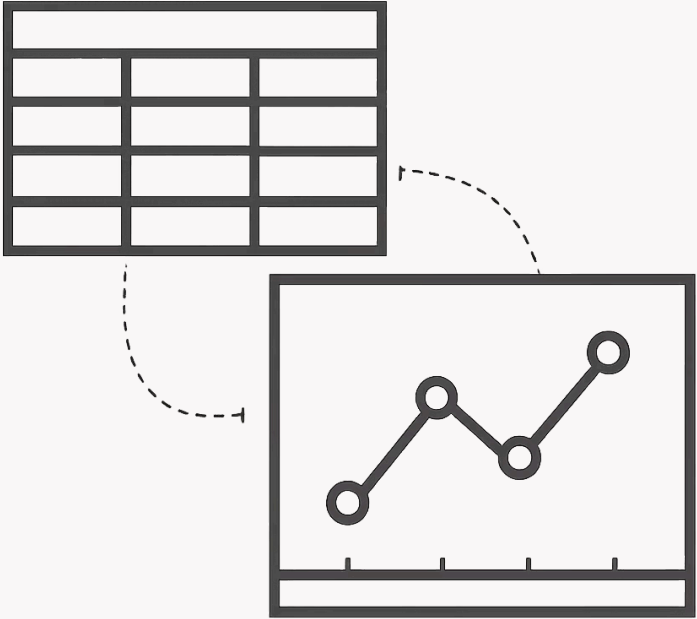
Age, Gender, Location,
Subscription

Purchase Details

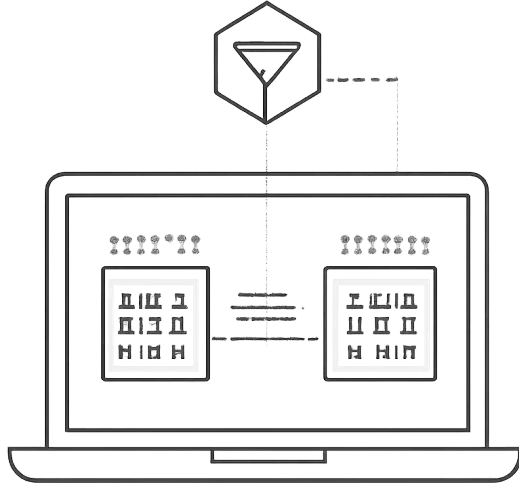
Item, Category, Amount,
Season

Behavior

Discount, Frequency, Review Rating



Data Preparation in Python



→ Load & Explore

pandas — `df.info()`, `.describe()`

→ Clean Missing Data

Imputed Review Rating by category median

→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ Load to MySQL

Cleaned DataFrame ready for SQL analysis

SQL Analysis — Revenue & Ratings

Revenue by Gender

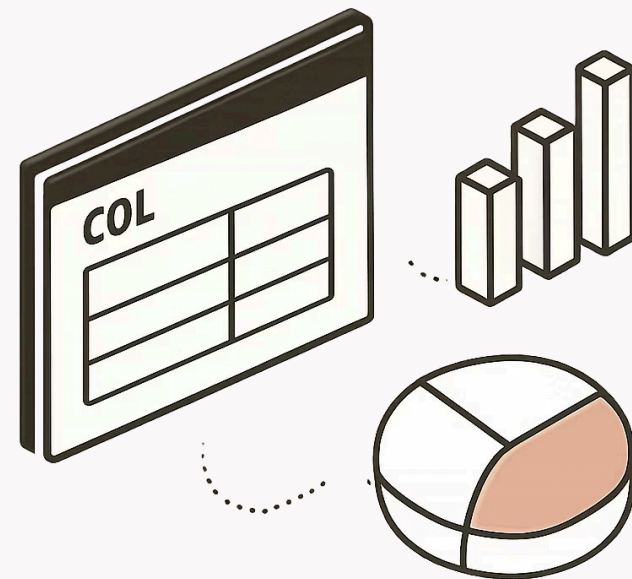
Male: **\$157,890** · Female: **\$75,191**

High-Spending Discount Users

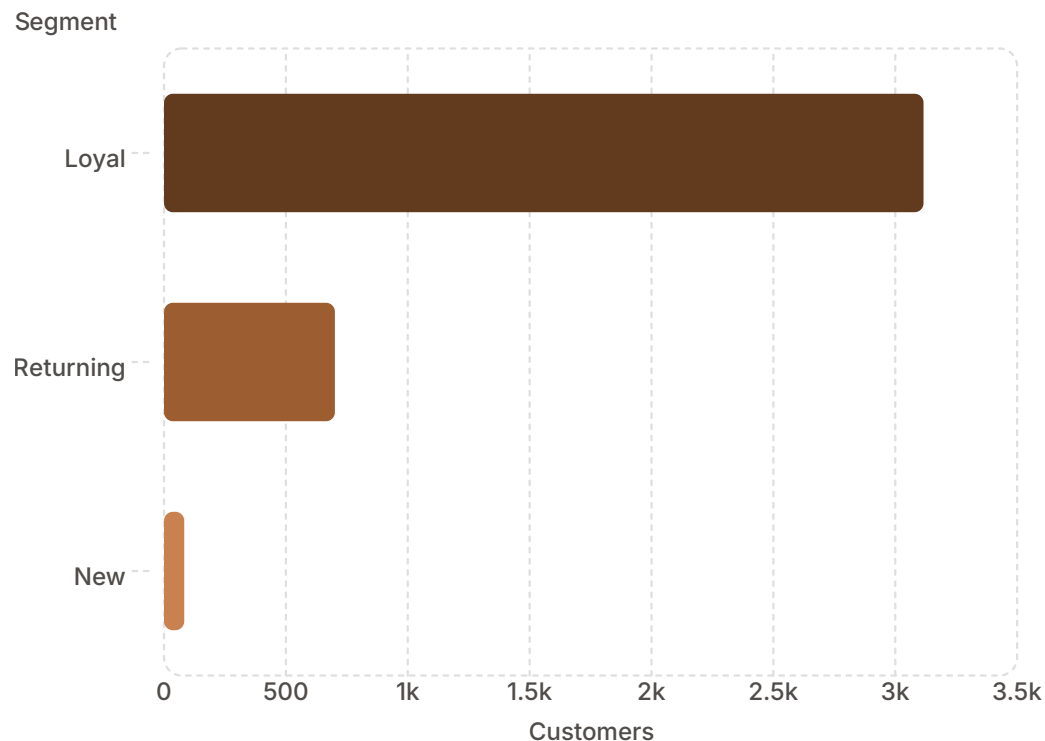
839 customers used discounts above average spend

Top 5 by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)



SQL Analysis — Behavior Insights



Subscribers vs Non-Subscribers

Non-subscribers: 2,847 customers, \$170,436 revenue

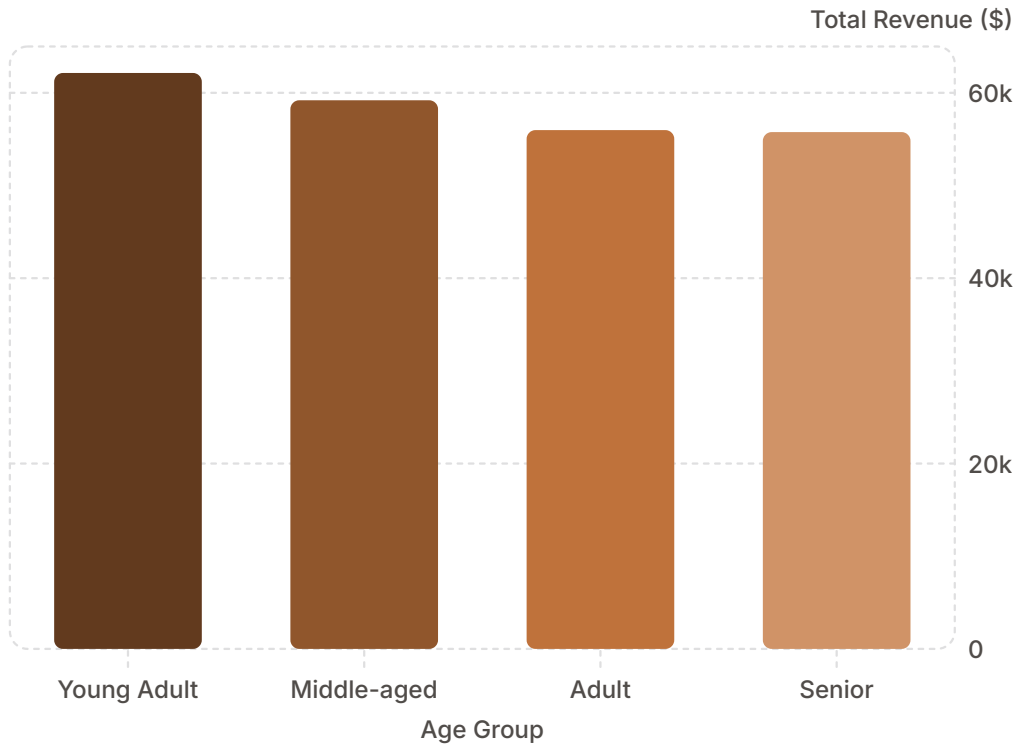
Discount-Heavy Products

Hat (50%), Sneakers (49.7%), Coat (49.1%)

Repeat Buyers & Subscriptions

958 subscribers vs 2,518 non-subscribers with 5+ purchases

Revenue by Age Group



Key Takeaway

Young Adults lead in revenue contribution at **\$62,143**. All age groups show relatively balanced spending.

i Shipping: Express buyers spend slightly more on average (\$60.48) vs Standard (\$58.46)

Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating



Subscription Split

27% subscribers · 73% non-subscribers



Revenue by Category

Clothing, Accessories, Footwear, Outerwear



Interactive Filters

Gender, Category, Shipping Type, Subscription



Business Recommendations

1

Boost Subscriptions

Promote exclusive subscriber benefits

2

Loyalty Programs

Reward repeat buyers → Loyal segment

3

Review Discount Policy

Balance sales lift with margin control

4

Product Positioning

Highlight top-rated, best-selling items

5

Targeted Marketing

Focus on high-revenue age groups & express-shipping users